MIL-B-49101A (CR)
11 September 1980
SUPERSEDING
MIL-B-49101 (EL)
16 December 1976

MILITARY SPECIFICATION

BUFFER, DATA TD-1065( )/G

This specification is approved for use by the Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Buffer, Data TD-1065( )/G which is a data transmission unit providing 6/12 channels of communication. It contains transmit and receive sections. The transmit section buffers and inserts 6/12 channels of digital data into a single serial output bit stream. The receive section removes and regenerates 6/12 channels of digital data simultaneously. (See 6.1 and 6.7).

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall be considered a superseding requirement.

SPECIFICATIONS

Military

MIL-P-116	Preservation-packaging, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-M-38510	Microcircuits, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications Research and Development Command, ATTN: DRDCO-CM, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-T-49102A(CR)	TEMPEST Requirements and Test for Buffer, Data TD-1065( )/G (CLASSIFIED)

Electronics Command

EL-CP0195-0001A	Nuclear, Survivability Requirements for Time Division Digital Multiplexer TD-1069( )/G(U) (CLASSIFIED)

STANDARDS

Military

MIL-STD-105	Standard Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461 Notice 4	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

Electronics Command

GL SM-B-327031	Gage List for TD-1065( )/G
DL SM-B-780110	Buffer, Data TD-1065( )/G
SM-A-780192	Test Requirement, Panel Assembly
SM-A-938578	19A6, Channel Unit
SM-A-780193	Test Requirement, Panel Assembly
SM-A-938579	19A4, Transmit Common

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SM-A-780194	Test Requirement, Panel Assembly
SM-A-938605	19A5, Receive Common
SM-A-780195	Test Requirement, Panel Assembly
SM-A-938606	19A3, Fault Locator
SM-A-938553	Test Requirement, Filter Assembly 19A7, Electrical
SM-A-780202	Test Requirement, Power Supply Assembly 19A1
SM-A-938550	Test Requirement, Filter Assembly 19A10 Audio

PUBLICATIONS

TM11-5805-382-12	Operator and Organizational Maintenance Manual-Multiplexer TD-660/G
SPI 1G0049	Special Packaging Instruction TD-1065( )/G

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be botained from the procuring activity or as directed by the Contracting Officer. Both the title and number should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First Article. Unless otherwise specified by the contract or purchase order, the contractor shall furnish First Article Units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of Drawing and Data List DL-SM-B-780110 including the requirements for parts, materials, and processes thereon.

3.3 Parts, Materials, and Processes; General. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 4.4).

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3.4 Electrical Characteristics.

3.4.1 Equipment Characteristics. (See 4.8.5). The Buffer, Data TD-1065( )/G shall insert and remove digital data from the 288/576 kb/s bit stream of Multiplexer TD-660( )/G. Audio signals applied to channels of the TD-1065( )/G shall be routed to the corresponding audio channels of the associated multiplexer for processing.

3.4.2 Channel Characteristics.

3.4.2.1 Impedance. (See 4.8.7) The input impedance of each channel shall be 600 ohms ±10 percent from 300 to 3500 Hz. The input impedance of each channel shall not exceed 750 ohms at 16,000 and 32,000 Hz. The output impedance of each channel shall be 600 ohms ±10 percent at 16 and 32 kilobits per second.

3.4.2.2 Frequency Window. (See 4.8.5) Channels of the TD-1065( )/G shall accept signals within a frequency window of ±0.72 Hz at 16 kb/s and ±1.44 Hz at 32 kb/s, and shall exhibit error free data transmission for a period of not less than 10 seconds.

3.4.2.3 Output Signal Levels. (See 4.8.5) The TD-1065( )/G shall provide channel output signals of 5.0 volts ±0.5 volts p-p when terminated in a 600 ±10 percent ohm load.

3.4.2.4 Signal Inversion. (See 4.8.5) The sense of signal applied to channels of TD-1065( )/G shall not be inverted at the output as a result of any internal processing.

3.4.2.5 Functional Capabilities. (See 4.8.5) Channels of the TD-1065( )/G shall provide the following functional capabilities:

a. Interface and process 16 kb/s subscribers on a 2-wire basis; interface and process 32 kb/s subscribers on a 4-wire basis.

b. Interface and process 16 kb/s and 32 kb/s subscribers on a 4-wire basis.

3.4.3 Inter-unit Signals. (See 4.8.8)

3.4.3.1 Transmit Side - Outputs from TD-1065( )/G.

3.4.3.1.1 XMT PCM OUT. The pulsewidth of the signal shall be 3.472 (±5.0 percent) microseconds for 6 channel signals or 1.736 (±5.0 percent) for 12 channel signals. The rise and fall times shall be 20 to 120 nanoseconds (ns). A digital "1" shall be a level between +0.4 and -0.4 volts and a digital "0" shall be a level between -2.2 and -2.6 volts. Ripple at either level shall be less than ±0.4 volts peak.

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3.4.3.1.2 12 CH TIM OUT. The frequency of the signal shall be 288.000 kHz ±0.003 percent for 6 channel operation or 576.000 kHz for 12 channel operation. The pulsewidth shall be 80 to 150 ns. The rise and fall times shall be 20 to 120 ns. The signal shall have a pulse peak 1.8 to 2.6 volts above a baseline of -2.0 to -2.3 volts. Ripple at the baseline shall be within ±0.5 volts.

3.4.3.1.3 AUX ADRS OUT. The signal shall be a negative going pulse that repeats at an 8 kHz rate. The pulsewidth shall be 3.48 (±5.0 percent) microseconds for 6 channel operation or 1.74 (±5.0 percent) microseconds for 12 channel operation. The base line of this signal is +0.0 to -0.4 volts and the negative peak is -2.2 to -2.6 volts. Rise and fall times shall be 20 to 120 ns. This signal shall be coincident with the address digit in the XMT PCM OUT train to within ±40 ns. Ripple at either level shall be within ±0.5 volts.

3.4.3.1.4 Phasing of XMT PCM OUT and 12 CH TIM OUT Signals. The leading edge of the 12 CH TIM OUT pulses shall lead the transitions in the XMT PCM OUT signal by the time ranges at the various temperatures as shown in the following phasing chart:

PHASING OF TIMING AND PCM SIGNALS

Temperature Range	Mode Number of Channels	Time Ranges (NS)
70° ± 20°F and High Temperatures (+125°F)	6 and 12	-20 to +120
Low Temperature (-25°F)	6	-80 to +120
	12	0 to +100

3.4.3.2 Receive Side - Outputs from TD-1065( )/G.

3.4.3.2.1 RCV PCM FR AUX. The pulsewidth of the signal shall be 3.472 (±5.0 percent) microseconds for 6 channel signals or 1.736 (±5.0 percent) microseconds for 12 channel signals. The rise and fall times shall be no greater than 120 ns. A digital "1" shall be a level between -0.0 and -0.4 volts and a digital "0" shall be a level between -2.2 and -2.6 volts. Ripple at either level shall be within ±0.5 volts.

3.4.3.2.2 RCVR AUX CLK OUT. The frequency of the signal shall be 288.000 kHz ±0.003 percent for 6 channel operation or 576.000 kHz ±0.003 percent for 12 channel operation. The pulsewidth shall be 80 to 150 ns. The rise and fall times shall be 20 to 120 ns. The leading edge of pulses shall lead the transitions in the RCV PCM FR AUX signal by 0 to 80 ns. The baseline of the signal shall be -2.2 to -2.6 volts and the pulse peak shall be -0.0 to -0.4 volts. Ripple at the baseline shall be within ±0.5 volts.

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3.4.3.2.3 RCVR AUX ADRS OUT. This signal shall be a positive going pulse that repeats at an 8 kHz rate. The pulsewidth shall be 80 to 150 ns. The rise and fall times shall be 20 to 120 ns. The leading edge of this pulse shall lead the leading edge of the frame pulse in the RCVR AUX CLK OUT signal by 0 to 160 ns. The baseline shall be -2.2 to -2.6 volts, and the pulse peak shall be +0.0 to -0.4 volts. Ripple at the baseline shall be within ±0.5 volts.

3.4.4 Fault Location. (See 4.8.5)

3.4.4.1 Test Lamp. A green light marked TEST shall be in conjunction with the TEST facility switch indicate the function (illuminated - ON) or malfunction (extinguished - OFF) of the following signals or equipment operations:

SWITCH POSITION	LAMP INDICATION
OFF	OFF
+10V	ON
-10V	ON
XMT PCM	ON
XMT TIM	ON
XMT FRAME	ON
RCV PCM	ON
RCV TIM	ON
RCV FRAME	ON
COM	ON
CHAN	ON

3.4.4.2 Alarms (Visible). A red light marked ALARM shall be illuminated upon the failure of any of the following signals, power supplies, and printed circuit card assemblies:

a. XMT PCM

b. XMT TIME

c. XMT FRAME

d. RCV PCM

e. RCV TIME

f. RCV FRAME

g. +10V Supply

h. -10V Supply

i. COMMON CARDS (COM)

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j. CHANNEL CARDS (CHAN)

With the service facility switch in the COM position a faulty common card shall be identified by an illuminated light emitting diode (LED) on the faulty card. With the service facility switch in the CHAN position, a faulty channel card shall be identified by an illuminated LED on the faulty card.

3.4.4.3 Alarms (Audible). An audible alarm shall operate upon the failure of the Transmit Common Card, the Receive Common Card, each Channel Card, and any interunit signal. A push-button switch shall turn off the audible alarm. When the equipment is restored to normal operation, after the audible alarm has been turned off, the audible alarm will again operate until the audible alarm switch is returned to the initial position. Operation of the alarm push-button switch shall turn off the audible alarm signal without delay.

3.5 Preconditioning.

3.5.1 Electrical Preconditioning. (See 4.3.2, 4.5, and 4.7.1) Prior to submission to the specified quality conformance inspections, each assembled TD-1065( )/G consisting of the chassis, plug-in panel assemblies and power supply plus spare assemblies defined by contract shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process.

3.5.2 Bounce Preconditioning. (See 4.3.2, 4.5 and 4.7.2) Each assembled TD-1065( )/G shall meet the requirements specified herein without subsequent processing after submission to the bounce preconditioning of 4.7.2. Each TD-1065( )/G shall be subjected to this process.

3.6 Service Conditions. (See 4.6)

3.6.1 Temperature. (See 4.6.1)

3.6.1.1 Low. (See 4.6.1.1) During and after testing as specified in paragraph 4.6.1.1, the equipment shall meet the requirements specified in paragraphs 3.4.1, 3.4.2, 3.4.3, and 3.4.4.

3.6.1.2 High. (See 4.6.1.2) During and after testing as specified in 4.6.1.2, the equipment shall meet the requirements specified in 3.4.1, 3.4.2, 3.4.3, and 3.4.4.

3.6.2 Humidity. (See 4.6.2) During and after testing as specified in 4.6.2, the equipment shall meet the requirements specified in paragraphs 3.4.1, 3.4.2, 3.4.3, and 3.4.4. At the conclusion of this test, the equipment shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials. Chassis joints shall not exhibit white salt deposits in excess of one trace of salt deposits per linear inch,

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not to exceed 50 percent of the brazed fillet width, exceed 3/16 inch in length nor cover a total area of 0.030 inch.

3.6.3 Low Pressure (Altitude). (See 4.6.3) During and after testing as specified in paragraph 4.6.3, the equipment shall meet the requirements specified in 3.4.1, 3.4.2, 3.4.3, and 3.4.4.

3.6.4 Vibration and Shock.

3.6.4.1 Vibration. (See 4.6.4) After testing as specified in paragraph 4.6.4, the equipment shall meet the requirements specified in 3.4.1, 3.4.2, 3.4.3, and 3.4.4.

3.6.4.2 Shock (Basic Design). (See 4.6.5) After testing as specified in paragraph 4.6.5, the equipment shall meet the requirements specified in 3.4.1, 3.4.2, 3.4.3, and 3.4.4.

3.6.4.3 Shock (Bench Handling). (See 4.6.6) After testing as specified in 4.6.6, the equipment shall meet the requirements specified in paragraphs 3.4.1, 3.4.2, 3.4.3, and 3.4.4.

3.6.4.4 Bounce, Loose Cargo. (See 4.6.7) After testing, as specified in 4.6.7, the equipment shall meet the requirements specified in 3.4.1, 3.4.2, 3.4.3, and 3.4.4.

3.6.5 Salt Fog. (See 4.6.8) After testing as specified in 4.6.8, the equipment shall meet the requirements specified in 3.4.1, 3.4.2, 3.4.3, and 3.4.4 and when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surfaces, or exfoliation (see 6.4.2) of metal.

3.6.6 Fungus. (See 4.6.9) After testing as specified in 4.6.9, the equipment shall be visually examined with the aid of a 10-power magnifier. The equipment shall exhibit no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.4.4) as evidenced by growth colonization (see 6.4.3) which includes branching (see 6.4.1) and sporulation (see 6.4.5) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.4.6) shall not be included in this evaluation.

3.7 TEMPEST. (See 4.14) The TD-1065( )/G shall meet the requirement of MIL-T-49102A(CR) when tested in accordance with 4.14.

3.8 Electromagnetic Interference (EMI). (See 4.10) The equipment shall meet the following EMI requirements of MIL-STD-461A, Notice 4:

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CE02	CS02	RE02	RS03
CE03	CS06	RE02.1	RS03.1
CE04			
CE05			

a. The frequency range for radiated emission (RE) requirements is 14 kHz to 1000 MHz.

b. The frequency range for radiated susceptibility (RS) requirements is 10 kHz to 1000 MHz.

3.8.1 Bonds and Grounds. (See 4.10) Bonding and grounding impedances shall not exceed 2.5 milliohms.

3.9 Finish. (See 4.4) The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings.

3.10 Marking. (See 4.4)

3.10.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.10.2 Visibility. Wherever practicable, parts and components shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.10.3 Serial Numbers. The Buffer, Data TD-1065( )/G shall have consecutively assigned serial numbers unless otherwise specified by the Contracting Officer.

3.11 Microelectronic Integrated Circuits. (See 4.4) All microelectronic integrated circuit devices used in this equipment shall meet the requirements of MIL-M-38510 for Class B devices or their individual drawing requirements.

3.12 Interchangeability. (See 4.12) Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454.

3.13 Panel Assemblies (Electrical Test). (See 4.9) The TD-1065( )/G panel assemblies listed below shall meet the requirements of their respective test requirement drawings as specified in 4.9.

Panel Assembly	Test Requirements
Power Supply 19A1	SM-A-780202
Panel Assembly 19A3	SM-A-780195
	*SM-A-938606
Panel Assembly 19A4	SM-A-780193

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Panel Assembly	Test Requirements
Panel Assembly 19A5	*SM-A-938579 SM-A-780194
Panel Assembly 19A6	*SM-A-938605 SM-A-780192
Filter Assembly 19A7	*SM-A-938578 SM-A-938553
Filter Assembly 19A10	SM-A-938550

*These drawings contain Automatic Test Equipment (ATE) acceptance procedures and programs.

3.14 Safety Engineering. (See 4.13)

3.14.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Radioactive Materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spare gaps, lenses).

3.15 Operational Test. (See 4.8.5) Each fabricated and tested TD-1065( )/G shall be subjected to a final operational test in accordance with paragraph 4.8.5 after all other inspections and tests are completed to determine that the TD-1065( )/G is operational prior to packaging.

3.16 Reliability. (See 4.11) The Buffer, Data TD-1065( )/G shall demonstrate a specified mean-time-between-failure (θ_o) of 2500 hours.

3.17 Nuclear Survivability. (See 4.15) Buffer, Data TD-1065( )/G shall withstand the nuclear environment as specified in EL-CPC195-0001A. The equipment shall meet the operational requirements of Section 3 herein, after being subjected to the specified nuclear environments. The equipment shall be energized during exposure.

3.18 Workmanship. (See 4.4) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268:

General requirements for plastic material and plastic parts.

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Wiring and cabling, including:
Slack
Protection
Clearance
Splicing
Connections, general
Grounding, general
Shielding on wire and cable
Soldering and brazing and welding
Controls
Self-Tapping screws
Printed Wiring Assemblies
Printed Wiring Boards

b. In this specification:

Construction
Safety Factors
Marking

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for Government Inspections shall be made available by the contractor.

4.2 Classification of Inspection. Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery (See 4.3).

b. Inspection covered by subsidiary documents (See 4.4).

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery (See 4.5).

(2) Quality conformance inspection of preparation for delivery (See 4.16).

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4.3 First Article. Unless otherwise specified in the contract or purchase order, the First Article Inspection shall be performed by the contractor.

4.3.1 First Article Units. The contractor shall furnish 13 First Article Units of the complete TD-1065( )/G.

4.3.2 First Article Inspection. The First Article Inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein except that all hardwired assemblies shall undergo inprocess Group A Inspection prior to electrical and bounce preconditioning.

4.3.3 First Article Data. The First Article Test Plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Parts, Materials, and Processes; General	3.3
Finish	3.9
Marking	3.10
Microelectronic Integrated Circuits	3.11
Workmanship	3.18

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. The contractor shall perform the inspections specified in 4.4, 4.7.1, 4.7.2, and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each unit (and each subassembly and assembly identified in Table II) on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. The order of inspection of the subassemblies, assemblies and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in Table II shall be subjected to a 100 percent visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in Table II.

b. After completion of the above 100 percent inspection by the contractor discrete lots shall be formed from the subassemblies and assemblies which passed the 100 percent inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's

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TABLE I

FIRST ARTICLE INSPECTION

Inspections	Rqt. Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6-13
1. Inspections covered by subsidiary documents		4.4	Inspection to be performed on all units					
2. Electrical preconditioning	3.5.1	4.7.1	Inspection to be performed on all units					
3. Bounce preconditioning	3.5.2	4.7.2	Inspection to be performed on all units					
4. Group A Inspection	(See Table II)		Inspection to be performed on all units					
5. Group B Inspection	(See Table III)		Inspection to be performed on all units					
6. TEMPEST	3.7	4.14	1					
7. EMI	3.8	4.10					1	
8. Nuclear Survivability	3.17	4.15					2	
9. Group C Inspection								
Low Temperature	3.6.1.1	4.6.1.1		2				
High Temperature	3.6.1.2	4.6.1.2		3				
Humidity	3.6.2	4.6.2			5			
Low Pressure (Altitude)	3.6.3	4.6.3		1				
Vibration	3.6.4.1	4.6.4			1			
Shock, Basic Design	3.6.4.2	4.6.5			2			
Shock, Bench Handling	3.6.4.3	4.6.6			3			
Bounce Loose Cargo	3.6.4.4	4.6.7			4			
Salt Fog (See Note C)	3.6.5	4.6.8		4				
Fungus (See Note C)	3.6.6	4.6.9				1		
10. Group D Inspection								
Reliability	3.16	4.11						1

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TABLE I

FIRST ARTICLE INSPECTION (Cont'd.)

NOTES:

a. The inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by Inspections 6, 7, 8, 9, and 10 specify the order of inspections for the indicated unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

indicated in Table II (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies from the discrete lots which have passed the above Quality Assurance Verification Inspection shall be assembled into the end equipments which shall then be subjected by the contractor to the preconditioning requirements of 4.7 and a 100 percent visual/mechanical and electrical inspection in accordance with Table II in that order.

NOTE: Assemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's 100 percent Group A Inspection of the end equipment.

d. After completion of the above 100 percent inspection by the contractor, discrete lots shall be formed from units which passed the 100 percent inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQL's indicated in Table II. Group A electrical inspection shall be performed in the order shown in Table II with the exception that the Operational Test (4.8.5) shall be last (See 6.6).

NOTE: Assemblies and subassemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's sampling inspection of the end equipment.

4.5.2 Group B Inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from

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TABLE II

GROUP A INSPECTION

Inspection	Rqt. Para.	Ref. Fig.	Insp. Para.	AQL Major	AQL Minor
Visual-Mechanical (Including spare assemblies)					
TD-1065( )/G Case and Covers	3.18		4.13	2.5*	10.0*
Power Supply 19A1	3.18		4.13	2.5*	4.0*
Panel Assembly 19A3	3.18		4.13	2.5*	4.0*
Panel Assembly 19A4	3.18		4.13	2.5*	4.0*
Panel Assembly 19A5	3.18		4.13	2.5*	4.0*
Panel Assembly 19A6	3.18		4.13	2.5*	4.0*
Filter Assembly 19A7	3.18		4.13	2.5*	4.0*
Filter Assembly 19A10	3.18		4.13	2.5*	4.0*
Electrical Performance (Including spare assemblies)					
Power Supply 19A1	3.13		4.9	1.0%	**
Panel Assembly 19A3	3.13		4.9	1.0%	**
Panel Assembly 19A4	3.13		4.9	1.0%	**
Panel Assembly 19A5	3.13		4.9	1.0%	**
Panel Assembly 19A6	3.13		4.9	1.0%	**
Filter Assembly 19A7	3.13		4.9	1.0%	**
Filter Assembly 19A10	3.13		4.9	1.0%	**
Electrical					
Operational	3.15	1, 1a	4.8.5	1.0%	**

*Defects per hundred units
**All electrical defects are major.

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units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B Sampling Plan. The Group B AQL shall be as shown in Table III and the inspection level shall be S-4.

4.5.2.2 Order of Inspection Within Group B . Group B inspection shall be performed in the order listed in Table III.

TABLE III

GROUP B INSPECTION

Inspection	Rqt. Para.	Ref. Fig.	Insp. Para.	AQL
Interchangeability				
Mechanical	3.12		4.12	6.5%
Channel Impedance	3.4.2.1	1,1a,3	4.8.7	4.0%
Interunit Signals	3.4.3		4.8.8	4.0%

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C Inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month. The unit shall be subjected to High and Low Temperature testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, three units shall be selected from the first production lot. For subsequent Group C inspection, three units shall be selected every fourth month. One unit shall be subjected to "TEMPEST" testing. One unit shall be subjected to EMI testing. One unit shall be subjected to Vibration; Shock, Basic Design; Shock, Bench Handling; and Bounce, Loose Cargo testing in that order.

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TABLE IV

GROUP C INSPECTION

Inspection	Rqt. Para.	Ref. Fig.	Insp. Para.
Subgroup 1			
High Temperature	3.6.1.2		4.6.1.2
Operational	3.15	1,1A	4.8.5
Interunit Signals	3.4.2.6	1,1A	4.8.8
Low Temperature	3.6.1.1		4.6.1.1
Operational	3.15	1,1A	4.8.5
Interunit Signals	3.4.2.6	1,1A	4.8.8
Subgroup 2			
TEMPEST	3.7		4.14
Vibration	3.6.4.1		4.6.4
Shock, Basic Design	3.6.4.2		4.6.5
Shock, Bench Handling	3.6.4.3		4.6.6
Bounce, Loose Cargo	3.6.4.4		4.6.7
EMI	3.8		4.10
Subgroup 3			
Humidity	3.6.2		4.6.2
Low Pressure (Altitude)	3.6.3		4.6.3
Salt Fog	3.6.5		4.6.8 (NOTE 1)
Fungus	3.6.6		4.6.9 (NOTE 1)

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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4.5.3.1.3 Subgroup 3. For this subgroup three units shall be selected from the first production lot. For subsequent Group C inspection, three units shall be selected every fifth month. One unit shall be subjected to Humidity, and Low Pressure (Altitude) testing in that order. One unit shall be subjected to Salt Fog testing. One unit shall be subjected to Fungus testing.

4.5.3.2 Order of Inspection Within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract.

4.5.4 Reinspection of Conforming Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group C Inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A Inspection only.

4.5.5 Group D Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on units from lots which have been subjected to and met Group A and Group C inspection. Test conditions, methods, and procedures shall be in accordance with paragraph 4.11.

TABLE V - GROUP D INSPECTION

Inspection	Req. Para.	Insp. Para.
Reliability	3.16	4.11

4.5.5.1 Sampling for Group D Inspection. Eight units shall be selected at random from the first month's production and subjected to the test of paragraph 4.11. For subsequent Group D inspections, eight units shall be selected at random from every sixth month's production.

4.5.5.2 Group D Failures. Actions required relative to Group D failures shall be as specified in the contract.

4.5.5.3 Reinspection of Conforming Group D Sample Units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Service Condition Tests.

4.6.1 Temperature Test.

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4.6.1.1 Low Temperature. (See 3.6.1.1) The equipment shall be tested in accordance with Test Method 502.1 Procedure I of MIL-STD-810. Storage temperature shall be minus 70 degrees F. Storage time shall be 24 hours. Operation temperature shall be minus 25 degrees F. At operation temperature, the test shall start within 5 minutes after power application to the unit. Performance tests, as specified in 4.8.6, shall be made during testing, as specified in Test Method 502.1. After testing, performance tests, as specified in 4.8.5 and 4.8.8, shall be made.

4.6.1.2 High Temperature. (See 3.6.1.2) The equipment shall be tested in accordance with Test Method 501.1, Procedure II of MIL-STD-810. Operating temperature shall be plus 125 degrees F. Performance tests, as specified in 4.8.6, shall be made during testing, as specified in Test Method 501.1. After testing, performance tests, as specified in 4.8.5 and 4.8.8, shall be made.

4.6.2 Humidity Test. (See 3.6.2) The equipment, with all covers removed, shall be tested in accordance with Test Method 507.1, Procedure II of MIL-STD-810. Performance testing as specified in 4.8.6 shall be made during the 24 to 28 hour period and the last 5 hour period of each 48 hour cycle. At the conclusion of testing, performance testing as specified in 4.8.5 shall be made. Failure of the equipment to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Low Pressure (Altitude) Test. (See 3.6.3) The equipment shall be tested in accordance with Test Method 500.1, Procedure I of MIL-STD-810. In Step 2, the rate of altitude change shall not exceed 2500 ±250 ft. per minute. Performance tests, as specified in 4.8.6, shall be made during testing, as specified in Test Method 500.1. After testing, performance tests, as specified in 4.8.5 shall be made.

4.6.4 Vibration Test. (See 3.6.4.1) The equipment shall be tested in accordance with Test Method 514.2, Procedure VIII of MIL-STD-810. Table 514.2-VI for equipment condition "wheeled vehicles" and Curve V of Figure 514.2-6 shall be used. Performance testing as specified in 4.8.5 shall be made after testing as specified in Test Method 514.2.

4.6.5 Shock (Basic Design) Test. (See 3.6.4.2) The equipment shall be tested in accordance with Test Method 516.2, Procedure I of MIL-STD-810. Performance testing as specified in 4.8.5 shall be made after testing as specified in Test Method 516.2, Figure 516.2-2 (peak value (A) g's-20, nominal duration (D)-11 ms).

4.6.6 Shock (Bench Handling) Test. (See 3.6.4.3) The equipment shall be tested in accordance with Test Method 516.2, Procedure V of MIL-STD-810. Performance testing as specified in 4.8.5 shall be made after testing as specified in Test Method 516.2.

4.6.7 Bounce, Loose Cargo Test. (See 3.6.4.4) The equipment shall be tested in accordance with Test Method 514.2, Procedure XI, Part 2 of MIL-STD-810. Performance testing, as specified in 4.8.5, shall be made after testing.

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4.6.8 Salt Fog Test. (See 3.6.5) The equipment shall be tested in accordance with Test Method 509.1 of MIL-STD-810. At the conclusion of the test the equipment shall be examined for evidence of visible corrosion. Performance testing as specified in 4.8.5 shall be made after testing as specified in Test Method 509.1. Failure of the equipment to meet the requirements of 3.6.5 shall constitute failure of this test.

4.6.9 Fungus Test. (See 3.6.6) The equipment, with all covers removed, shall be tested in accordance with Test Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization (See 6.4.3) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment together with cases and accessories, is permitted for 72 hours prior to the start of fungus test. Failure of the equipment to meet the requirements of 3.6.6 shall constitute failure of this test.

4.7 Preconditioning. (See 3.5).

4.7.1 Electrical Preconditioning. (See 3.5.1) Electrical preconditioning for the equipment and spare assemblies shall meet the requirement of 3.5.1.

4.7.1.1 Test Details.

a. The equipment shall be mounted on a vibration table in its normal operating position. Test Level B of MIL-STD-781 shall apply for Temperature and Vibration requirements. The equipment shall be connected as shown in Figure 1 or 1a. The equipment shall be turned on and operated (all channels processing data) for 24 hours and then turned off for a maximum of 1 hour followed by a second 24 hour operating period.

b. Prior to the start and at the end of each 24 hour operating period, the equipment shall be subjected to the test specified in 4.8.5.

c. Test data shall be recorded at the end of each operating period.

d. The chassis, plug-in printed circuit panel assemblies and power supplies shall be considered to have been successfully electrically preconditioned when they have individually passed a cumulative 48 hour operating preconditioning period without a failure during the last 24 hour period. If during the last 24 hour operating preconditioning period a plug-in panel assembly, power supply, or chassis fails, it may be replaced with a preconditioned panel, power supply, or chassis and the preconditioning continued for a 24 hour failure-free period. The restart of a new 48 hour preconditioning period is not required. A plug-in panel assembly, power supply or chassis failing operating preconditioning shall, after repair, be resubmitted to and pass the entire 48 hour operating preconditioning period prior to final electrical assembly tests by the contractor.

e. The unit chassis shall have no less than 48 hours of operational preconditioning nor more than 168 hours of operational preconditioning (exclusive

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of production test time) when used as a vehicle for preconditioning power supplies and plug-in assemblies (including all spares).

f. Assemblies failing electrical acceptance tests subsequent to satisfactory completion of electrical preconditioning may be repaired with an electrically preconditioned assembly and electrically retested, accepted and delivered without repeat of electrical preconditioning.

4.7.1.2 Marking of Electrically Preconditioned Units. All electrically preconditioned plug-in panel assemblies, power supplies and chassis shall be physically identified by a preconditioning stamp. The stamp shall be affixed to the front panel near the nameplate of each unit which has successfully completed the preconditioning as outlined herein. The stamp on preconditioned plug-in panel assemblies shall be affixed to the bottom side, and to the bottom outside edge of the power supply subassembly 19A1. The stamp shall be a "p" in a circle and shall be marked on the preconditioned unit per the Group III requirements of MIL-M-13231.

4.7.2 Bounce Precondition. (See 3.5.2) Each unit, in its non-operating condition, shall be placed in its normal operating position on the table of the package tester Type 1000 SC as made by the L.A.B. Corporation, Skaneatecles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. Each unit of each lot shall be subjected to this preconditioning. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.8 Electrical Tests.

4.8.1 Standard Conditions. Unless otherwise specified, all tests shall be performed at 115 volts ±5 percent from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperatures, pressure and humidity per MIL-STD-810 unless otherwise specified.

4.8.2 Adjustment for Normal Operation (TD-1065( )/G).

a. Apply power to the equipment.

b. Adjust all channel mode switches for 4-wire operation.

4.8.2.1 Alignment for Normal Operation (TD-660( )/G).

a. Apply power to the equipment.

b. Adjust equipment for 12 channel operation in accordance with TM 11-5805-383-12.

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c. Align audio channels for 4-wire operation (-4 dBm in and -4 dBm out) using internal test tone oscillator and test align meter in accordance with TM 11-5805-383-12.

4.8.2.2 Adjustment for Normal Operation (Mux Simulator).

a. Apply power to the equipment.

b. Operate in 12 channel mode.

4.8.3 Standard Test Equipment. Standard test equipment is indicated in test figures by manufacturer and model number. In all cases, equivalent substitutes may be used. The equipment indicated in the figures is listed in Table VI. Standard test equipment shall be provided by the contractor. Government approval is required prior to use of proposed substitute items.

TABLE VI

STANDARD TEST EQUIPMENT

Designation	Function	Manufacturer
HP 200CD	Wide Range Oscillator	Hewlett-Packard Co.
HP 400D	Vacuum Tube Voltmeter	Hewlett-Packard Co.
TEK 545A	Oscilloscope	Tektronix, Inc.

4.8.4 Special Test Equipment. Unless otherwise specified, special test equipment shall be provided by the contractor subject to approval of the Government.

4.8.5 Operational Test.

a. Connect the equipment per Figure 1 or 1a.

b. Align TD-660( )/G or adjust mux simulator for normal operation.

c. Adjust TD-1065( )/G for normal operation.

d. Check all indicators on the TD-1065( )/G for the proper indication.

e. The requirements of 3.4.4.1 shall be met.

f. Turn off the AC power to the associated TD-660( )/G or mux simulator unit.

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g. For this condition, note the following indications on the TD-1065( )/G unit:

(1) The red ALARM lamp illuminates and the audible alarm sounds.

(2) The red LED's on the 19A4 and 19A5 panels illuminate.

(3) Audible alarm is silenced utilizing push button switch.

h. For this condition also note the status of the green TEST lamp on the TD-1065( )/G unit for the TEST switch positions listed below:

SWITCH POSITION	LAMP INDICATION
OFF	OFF
+10V	ON
-10V	ON
XMT PCM	OFF
XMT TIM	OFF
XMT FRAME	OFF
RCV PCM	OFF
RCV TIM	OFF
RCV FRAME	OFF
COM	OFF
CHAN	ON

i. Turn on the AC power to the associated TD-660( )/G or mux simulator unit.

j. For this condition, note the following indications on the TD-1065( )/G unit:

(1) The red ALARM lamp extinguishes and the audible alarm sounds.

(2) The red LED's on the 19A4 and 19A5 panel extinguish.

(3) The audible alarm is silenced utilizing the push button switch.

(4) All indicators provide the proper indications.

k. The requirements of 3.4.4.1, 3.4.4.2, and 3.4.4.3 shall be met.

l. With the TEST facility switch in the CHAN position, apply a data source operating at 9.6 kb/s to each channel one at a time.

m. For each application, note the following:

(1) The red ALARM lamp illuminates.

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(2) The green TEST lamp extinguishes.

(3) The audible alarm sounds.

(4) Each channel's LED illuminates.

n. The requirements of 3.4.4.1, 3.4.4.2, and 3.4.4.3 shall be met.

o. Utilizing TA-312 Telephones talk and listen on each channel one at a time. (Delete when utilizing mux simulator).

p. The voice communications requirement of 3.4.1 shall be met. (Delete when utilizing mux simulator).

q. Apply a data source operating at 16 kb/s to channel 1.

r. Correlate data transmission for errors (not less than 10 seconds.)

s. Repeat steps q. and r. above for all remaining channels.

t. Apply a data source operating at 32 kb/s to channel 1.

u. Repeat step r. above for all remaining channels.

v. The requirements of 3.4.2.2 and 3.4.2.4 shall be met.

w. Set all channel mode switches to "2w".

x. Repeat steps q, r, and s above.

y. Verify that data transmission is inhibited.

z. The requirements of 3.4.2.5 shall be met.

aa. Connect equipment per Figure 3.

ab. Measure the output signal level of channel 1. (32 kb/s data source applied).

ac. Repeat step x. above for all remaining channels.

ad. The requirements of 3.4.2.3 shall be met.

ae. Connect the equipment per Figure 2 or 2a.

af. Operate TD-660( )/G or mux simulator in six channel mode.

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ag. Repeat steps d. through ad. above (Odd numbered channels only).

4.8.6 Performance Test (Service Condition/Operating).

a. Connect equipment per Figure 1 or 1a.

b. Disconnect termination on AUX ADRS OUT connector on TD-1065( )/G.

c. Reconnect cable assembly with tee adapter UG-274/U or equivalent and termination.

d. Disconnect cable assemblies from PCM IN and TIM IN connectors on TD-660( )/G or mux simulator.

e. Connect tee adapters to the PCM IN and TIM IN connectors and reconnect cable assemblies.

f. Align TD-660( )/G or adjust mux simulator for normal operation.

g. Adjust TD-1065( )/G for normal operation.

h. Check all indicators on the TD-1065( )/G for proper indications.

i. The requirements of 3.4.4.1 shall be met. (Where access to switch functions are not practical during service condition tests, the absence of alarm function as defined in 3.4.4.3 shall be used to determine proper operation.)

j. Interconnect channels 1 through 12 in tandem.

k. Apply a data source operating 32 kb/s to the input of channel 1.

l. Apply a data correlator to the output of channel 12.

m. Correlate data transmissions for errors (not less than 10 seconds).

n. The requirements of 3.4.2.2 shall be met.

o. Measure the XMT PCM OUT, 12 CH TIM OUT, and AUX ADRS OUT signals at the tee adapters for upper and lower voltage levels, ripple, rise and fall times, pulsewidth, and phasing.

p. The requirements of 3.4.3.1 shall be met.

q. Connect the equipment per Figure 2 or 2a.

r. Operate the TD-660( )/G or mux simulator in the 6 channel mode.

s. Repeat Steps h. through p. above (Interconnect channels 1, 3, 5, 7, 9 and 11 in tandem).

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4.8.7 Channel Impedance Test.

a. Connect equipment (high speed side) per Figure 1 or 1a.

b. Connect equipment (low speed side) per Figure 4.

c. Align TD-660( )/G or adjust mux simulator for normal operation.

d. Adjust TD-1065( )/G for normal operation.

e. Check all indicators on the TD-1065( )/G for proper indications.

f. The requirements of 3.4.4.1 shall be met.

g. Measure the input impedance of channel 1 at 300, 1100, 3500, 16,000, and 32,000 Hz.

h. Apply a data source operating at 16 kb/s to the input of channel 1 and measure the output impedance of channel 1.

i. Apply a data source operating at 32 kb/s to the input of channel 1 and measure the output impedance of channel 1.

j. Repeat steps g. through i. above for channels 5 and 9.

k. The requirements of 3.4.2.1 shall be met.

4.8.8 Interunit Signals Test.

a. Connect equipment per Figure 1 or 1a.

b. Disconnect cable assemblies from XMT PCM OUT and 12 CH TIM OUT connectors on TD-1065( )/G.

c. Connect tee adapters UG-274/U or equivalent to the XMT PCM OUT and 12 CH TIM OUT connector and reconnect cable assemblies.

d. Align TD-660( )/G or adjust mux simulator for normal operation.

e. Adjust TD-1065( )/G for normal operation.

f. Check all indicators on the TD-1065( )/G for proper indications.

g. The requirements of paragraph 3.4.4.1 shall be met.

h. Measure the XMT PCM OUT, 12 CH TIM OUT, and AUX ADRS OUT signals at the connectors on the TD-1065( )/G for upper and lower voltage levels, ripple, rise and fall times, pulsewidth, and phasing.

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i. The requirements of paragraph 3.4.3.1 shall be met.

j. Measure the RCV PCM FR AUX, RCVR AUX CLK OUT, and RCVR AUX ADRS OUT signals at the connectors on the TD-1065( )/G for upper and lower voltage levels, ripple, rise and fall times, pulsewidth, and phasing.

k. The requirements of paragraph 3.4.3.2 shall be met.

l. Connect the equipment per Figure 2 or 2a.

m. Operate the TD-660( )/G or mux simulator in the 6 channel mode.

n. Repeat Steps f. through k. above.

4.9 Electrical Subassembly Test. (See 3.13) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings. Drawings SM-A-938606, SM-A-938579, SM-A-938605, and SM-A-938570 may be used to test A3, A4, A5, and A6 subassemblies. These drawings represent automated test equipment (ATE) approaches which are compatible with requirements of para 3.13. Deviations from these ATE procedures or identification of new ATE procedures is subject to prior approval of the procuring activity.

4.10 Electromagnetic Interference Test. (See 3.8)

4.10.1 First Article Electromagnetic Interference Test. The equipment shall be tested for compliance to the requirements of paragraph 3.8. Test setups and procedures shall be in accordance with the requirements of MIL-STD-462, Notice 3 as implemented by a contractor prepared, government approved, test plan. Prior to the start of EMI tests, compliance with the requirements of 3.8.1 shall be verified.

4.10.2 Quality Conformance Electromagnetic Interference Test. The equipment shall be tested for compliance to the following requirements:

CE 04	CS 02	RE 02	RS 03
	CS 06	RE 02.1	

Test setups and procedures shall be in accordance with the requirements of MIL-STD-462, Notice 3 as implemented by a contractor prepared, government approved, test plan. Prior to the start of EMI tests, compliance with the requirements of 3.8.1 shall be verified.

4.11 Reliability. (See 3.16)

4.11.1 First Article Reliability Test. The First Article Reliability inspection testing shall be in accordance with Test Plan IV of MIL-STD-781.

4.11.2 Quality Conformance (Group D) Reliability Test. The reliability test for Group D inspection shall be in accordance with Test Plan IVa of MIL-STD-781.

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4.11.3 Reliability Test Conditions. The detail test conditions procedures and methods for the tests described in 4.11.1 and 4.11.2 shall be in accordance with contractor prepared, Government approved Test Plan for TD-1065( )/G.

4.11.4 Reliability Test Conditions.

4.11.4.1 Performance Parameters to be Measured. Performance testing as specified in paragraph 4.8.5 shall be performed on all units subjected to Reliability testing.

4.11.4.2 Environmental and Operational Stresses. Test level B of MIL-STD-781 shall apply except as noted below:

a. Input voltage cycling specified in Test Level B does not apply.

b. Equipment ON-OFF cycling is revised to read "Equipment ON-OFF Cycling. The equipment shall be turned on and operated with all channels processing data for 3 hours, 55 minutes then turned off for 5 minutes. This cycle shall continue throughout the test."

c. The Operational Test of 4.8.5 shall be performed during the "ON" period of every sixth cycle. The time period expended in the performance of the operational test shall not exceed the "ON" time.

4.11.5 Vibration. During testing, the equipment shall be mounted on a vibration table positioned in its normal operating plane with all covers in place. Vibration shall be in a plane perpendicular to the 19A6 printed circuit cards. Covers may be removed as necessary during test periods, but must be replaced at the conclusion of the electrical test period.

4.11.6 Failure Reporting and Analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11.7 Failure Definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below performance limits specified in 4.8.5.

4.11.8 Accept-Reject Criteria. The accept-reject criteria shall be Test Plan IV or IVa of MIL-STD-781.

4.12 Inspection for Dimensional Interchangeability. (See 3.12) The gage points identified on GL SM-B-327031 shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.12.

4.13 Visual and Mechanical Inspection. (See 3.14 and 3.18) Equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be

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performed to verify compliance with those portions of 3.14 (safety engineering) which can be determined visually.

4.14 TEMPEST. (See 3.7) The equipment shall be tested as specified in MIL-T-49102A(CR).

4.15 Nuclear Tests. (See 3.17) Nuclear tests (Blast, EMP, Initial Nuclear Radiation) shall be performed on the equipment in accordance with paragraphs 4.15.1 through 4.15.3 inclusive.

4.15.1 Nuclear-Air Blast Test. The equipment shall be placed in a nuclear-air blast simulation facility and exposed to the air blast environment as specified in paragraph 3.17 Nuclear Survivability. The equipment shall meet the requirements of paragraph 3.4.1 herein following exposure. Typical test facilities for this test are located at:

a. BRL, Aberdeen, Maryland.

b. Blast Load Generator, Waterways Experimental Station, Vicksburg, Mississippi.

c. US Naval Civil Engrg Lab, Port Hueneme, California.

d. Sandia Corp, Albuquerque, New Mexico.

e. Air Force Weapons Lab, Albuquerque, New Mexico.

Calibration certification of listed test facilities or their equivalent are required by the COTR, 45 days prior to nuclear test. Verification of the wave shape of the test simulator is required. Calibration and wave shape verification shall be supplied to the COTR by the contractor.

4.15.2 Electromagnetic Pulse. The equipment shall be placed within a long wire electromagnetic pulse facility or its equivalent with connecting cables extended parallel to the electric field and perpendicular to the magnetic field; all individual units under the long wire, but widely separated. The equipment shall be exposed to the maximum output of the test facility in the above orientation and in each operating mode of the equipment. The equipment shall meet the requirements of paragraph 3.4.1 following exposure. Limits below the capabilities of the suggested facilities are not acceptable. Test data to satisfy the EMP requirements specified herein (paragraph 3.17, Nuclear Survivability) may be assembled by extrapolation to threat level and injection pulsing at the calculated level. Typical test facilities for this test are located at:

a. Martin Marietta Company, Orlando, Florida

b. Sandia Corporation, Albuquerque, New Mexico

c. Air Force Weapons Lab, Albuquerque, New Mexico

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d. HDL EMP Simulator, Woodbridge, Virginia

Calibration certification of listed test facilities or their equivalents are required by the COTR 45 days prior to nuclear test. Verification of the wave shape of the test facility is required. Calibration and wave shape verification shall be supplied to the COTR by the contractor.

4.15.3 Initial Nuclear Radiation Test. One complete set of equipment will be exposed to a LINAC and a pulsed or fast burst reactor in such a manner that all electronic piece parts and circuits in the set receive the neutron and gamma doses and rates specified in paragraph 3.17, Nuclear Survivability. The equipment shall meet the requirements of paragraph 3.4.1 herein following exposure. Typical test facilities for Initial Nuclear Radiation Tests are located at:

a. Fast Burst Reactors, White Sands, New Mexico

b. Sandia Pulsed Reactor, Albuquerque, New Mexico

c. Diamond Ordnance Radiation Facility, Washington, DC

d. Gulf General Atomic, LaJolla, California

e. Northrop Corp., Hawthorne, California

f. LINAC, WSMR, White Sands, New Mexico

g. Aurora, HDL, Adelphi, Maryland

h. HERMES II, Sandia Labs, Albuquerque, New Mexico

Calibration certification of listed test facilities or their equivalents are required by the COTR, 45 days prior to nuclear test. Calibration and wave shape verification shall be supplied to the COTR by the contractor.

4.16 Quality Conformance Inspection of Preparation for Delivery. Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with SPI 1G0049.

6. NOTES

6.1 Intended Use. Buffer, Data TD-1065( )/G operates in shelter assemblages of the Army Tactical Communications Systems (ATACS). The unit provides a 12 channel digital transmission capability when used in conjunction with Multiplexers TD-352/U, TD-660A/G, TD-660B/G and associated equipment. Interoperation with Multiplexer TD-660/G is precluded.

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6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, Level B or commercial preservation and packing (See Section 5).

c. When rough handling test is required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required, test tools and running spares.

g. First Article Inspection. (See 3.1)

h. Length of cables (if any are required).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Definitions.

6.4.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.4.2 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4.3 Growth Colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be the one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.4.4 Microbial Growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.4.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.4.6 Tubular Germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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6.5 Group C Inspection. (See paragraph 4.5.3.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the Contracting Officer on the adequacy of corrective action.

6.5.1 Initial Shipments. (See paragraph 4.5.3.3) Initial shipments of end items will be contingent only upon successful completion of all Group C tests.

6.6 Location of Operational Inspection. (See paragraph 4.8.5) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.7 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TD-1065H/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.8 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN

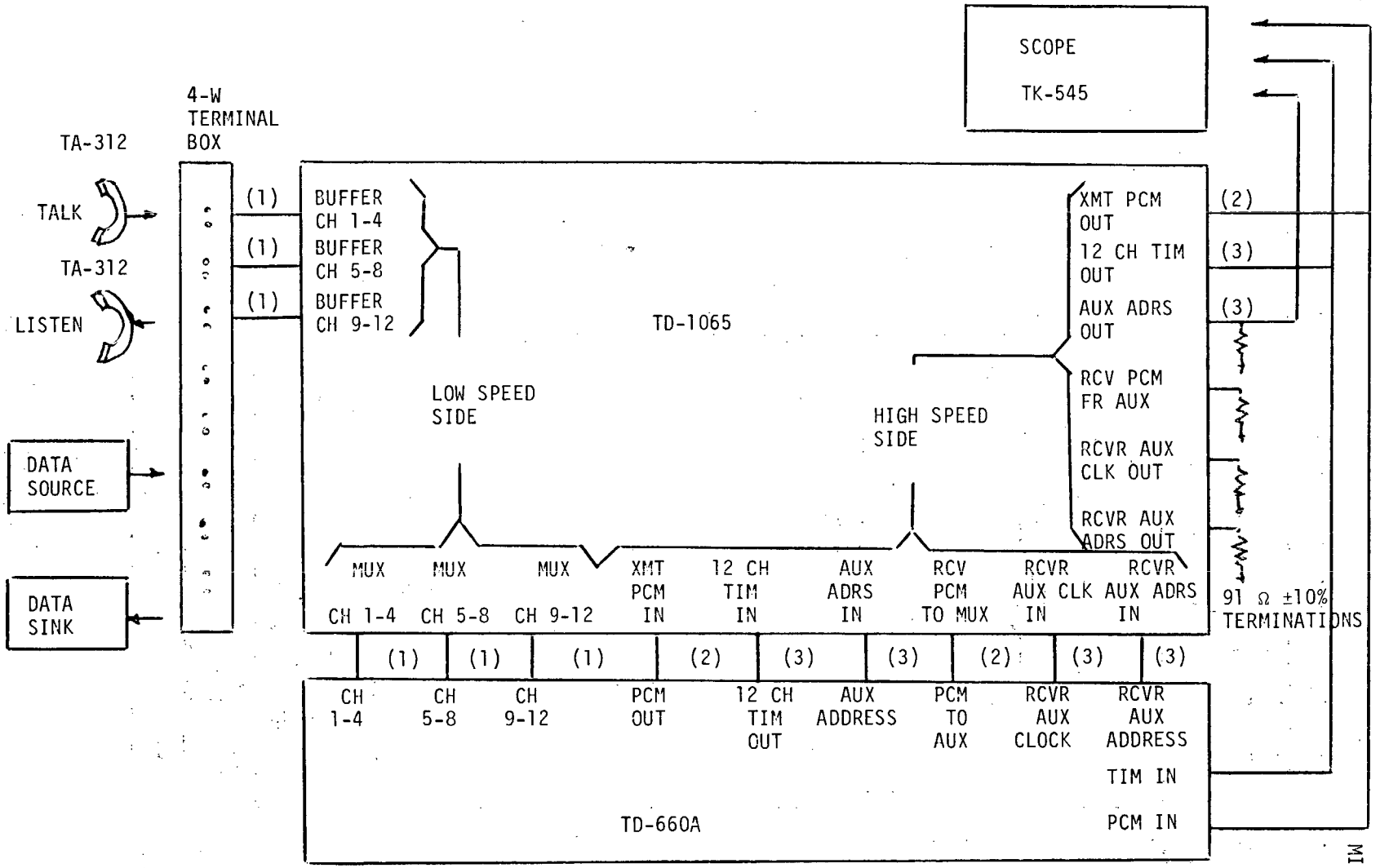
ARMY - CR

PREPARING ACTIVITY

ARMY - CR

Project 5805-A269

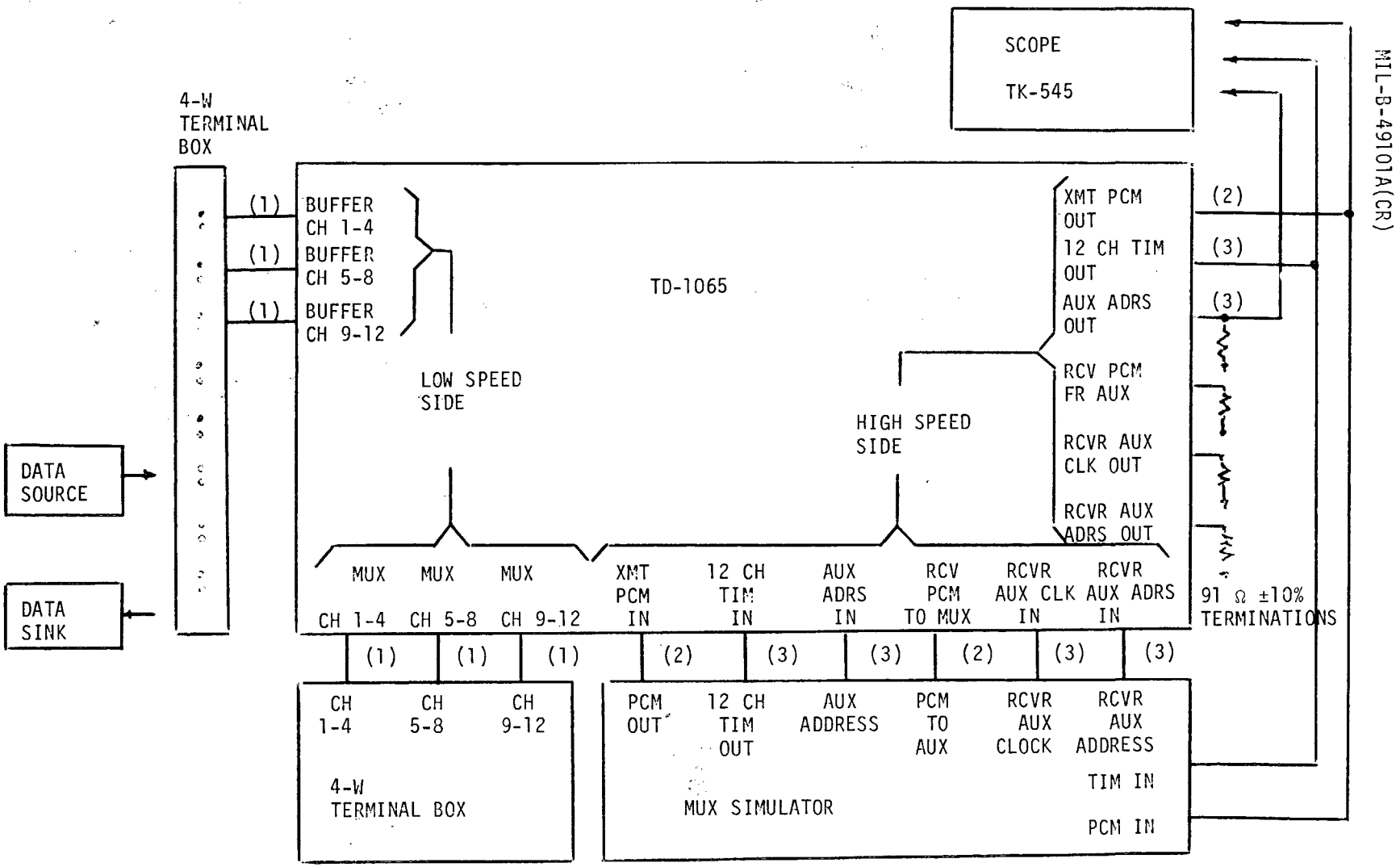
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(1) - CX-7870( )/TCC
(2) - CG-3473( )/U
(3) - CG-3419( )/U

FIGURE 1. OPERATIONAL CHECK

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(1) - CX-7870( )/TCC
(2) - CG-3473( )/U
(3) - CG-3419( )/U

FIGURE 1A - OPERATIONAL CHECK

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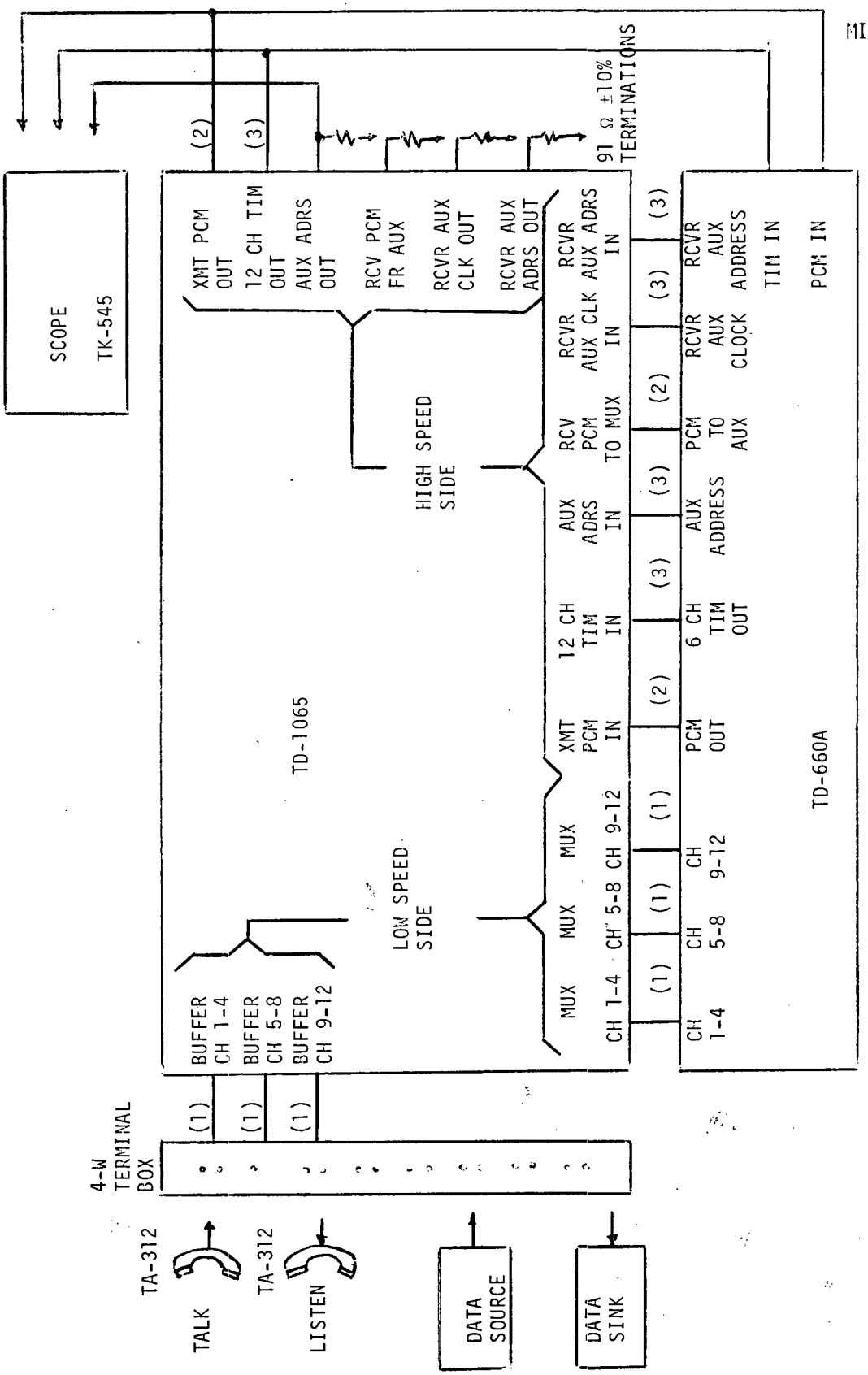


FIGURE 2. OPERATIONAL CHECK

(1) - CX-7870( )/TCC
(2) - CG-3473( )/U
(3) - CG-3419( )/U

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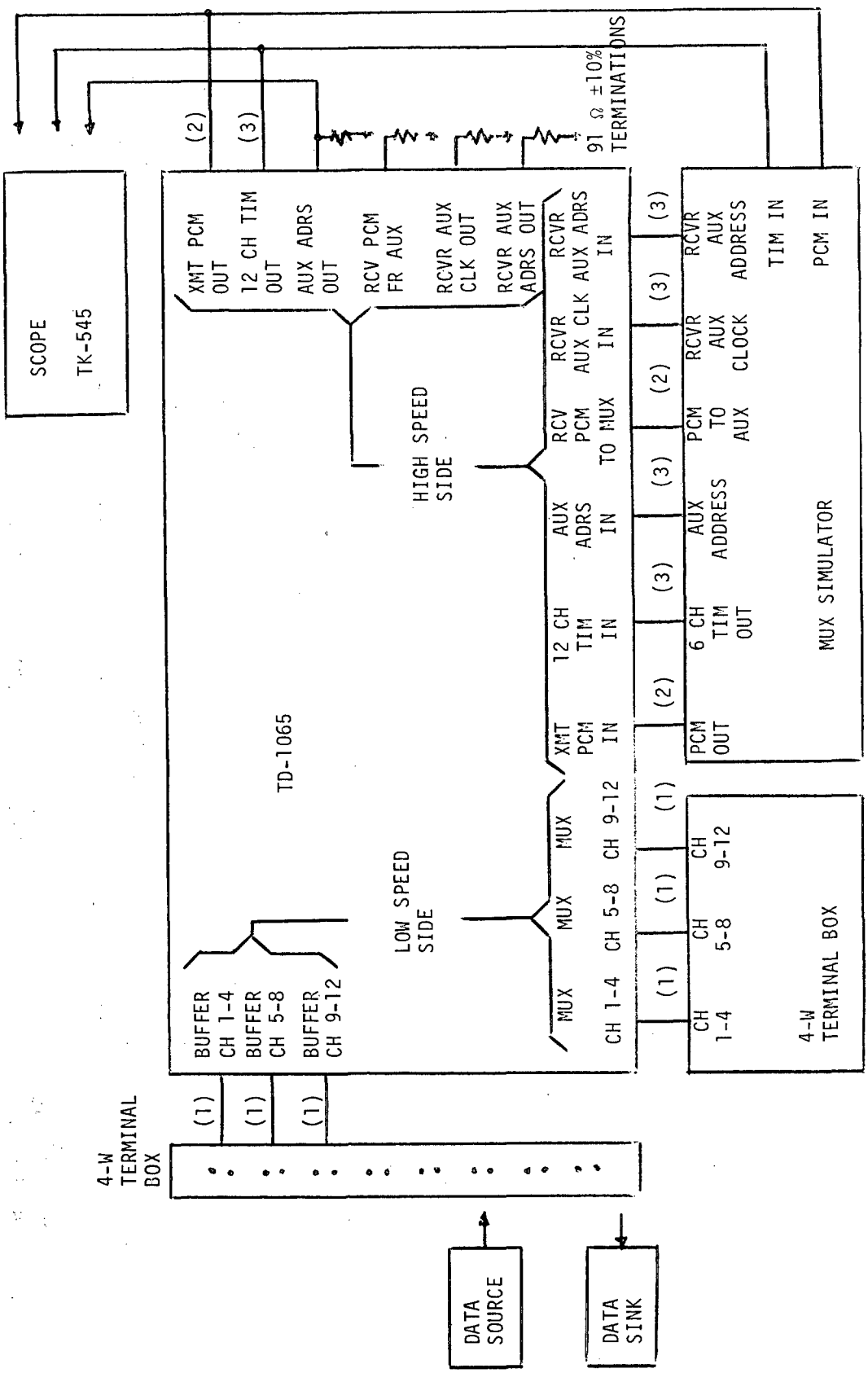


FIGURE 2A. OPERATIONAL CHECK

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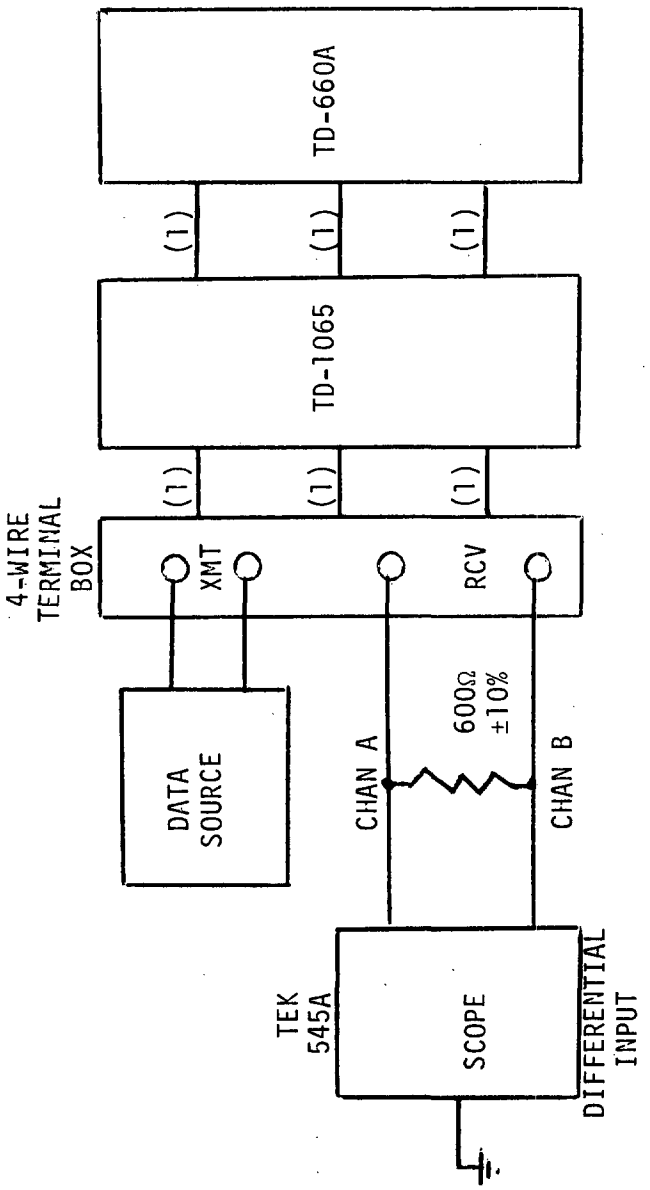


FIGURE 3. OUTPUT SIGNAL

(1) CX-7870( )/TCC

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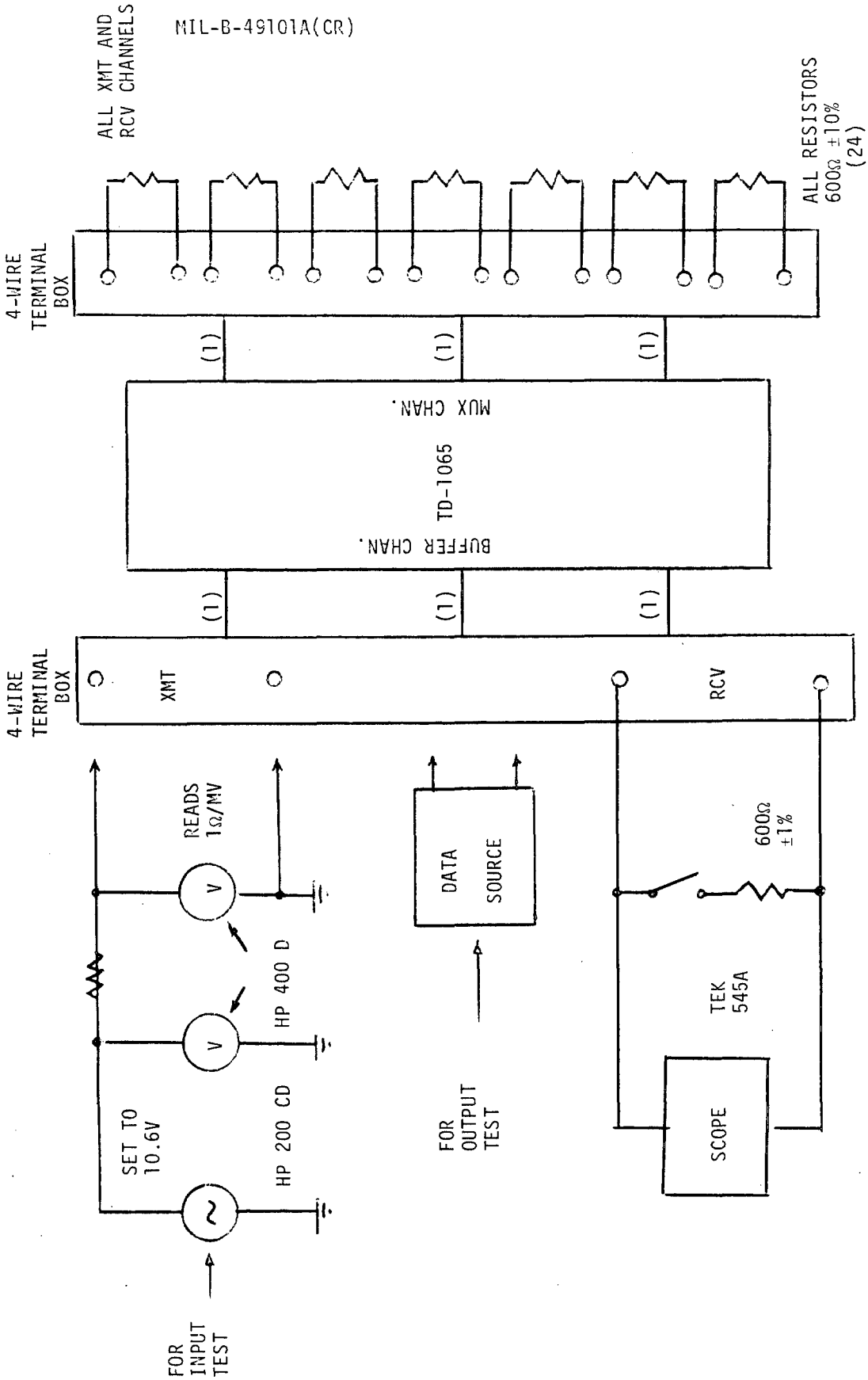


FIGURE 4. CHANNEL IMPEDANCE

(1) CX-7870( )/TCC

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